NATIONAL HIGHER SCHOOL OF MATHEMATICS

المدرسة الوطنية العليا في الرياضيات

Preparatory Cycle **Academic Year :** 2024/2025

First Year **Course : Algebra 1**

Worksheet 1 :
Mathematical logic and mathematical reasoning

The starred exercises are to be done in tutorial sessions

Exercise 1. *

1. *Give two statements P and Q such that*
   (a) $P \implies Q$ *is true and* $Q \implies P$ *is true.*
   (b) $P \implies Q$ *is true and* $Q \implies P$ *is false.*
   (c) $P \implies Q$ *is false and* $Q \implies P$ *true.*
2. *Is it possible to find two statements P and Q such that $P \implies Q$ is false and $Q \implies P$ is false ?*

Exercise 2. *

Let P, Q, and R be three statements.

1. *Show that the following compound statements are logically equivalent by using two different methods.*
   (a) $P \implies (Q \vee R)$ *and* $(P \wedge \overline{R}) \implies Q$.
   (b) $\overline{(P \vee \overline{Q})} \vee (\overline{P} \wedge \overline{Q})$ *and* $\overline{P}$.
   (c) $(P \wedge \overline{(\overline{P} \vee Q)}) \vee (P \wedge Q)$ *and* P.
2. *Without using the truth table, simplify the statement* $(\overline{P} \wedge Q) \vee (\overline{P} \wedge \overline{Q}) \vee (P \wedge Q)$.
3. *Verify whether the following compound statements are tautologies, contradictions, or contingencies.*
   (a) $(P \wedge Q) \wedge \overline{(P \vee Q)}$.
   (b) $(P \vee Q) \wedge \overline{P}) \implies Q$.
   (c) $(P \implies Q) \iff (\overline{P} \implies Q)$.
   (d) $((P \implies Q) \wedge (Q \implies R)) \implies (P \implies R)$.

Exercise 3.

Give the logical relationships between the following statements.

1. P_1 : *All men are mortal.*
2. P_2 : *All men are immortal.*
3. P_3 : *No man is mortal.*
4. P_4 : *No man is immortal.*
5. P_5 : *There are immortal men.*
6. P_6 : *There are mortal men.*

Exercise 4. *

1. *Let $f : \mathbb{R} \to \mathbb{R}$ be a map. Using the syntax of the predicates write the following sentences and then give their negation.*

(a) *f is positive.*
(b) *f is the zero function.*
(c) *f is even.*
(d) *f is odd.*
(e) *f is decreasing.*
(f) *f is strictly monotonic.*
(g) *f changes the sign.*
(h) *f is upper bounded.*
(i) *f is constant.*
(j) *f is the identity map.*
(k) *f is periodic.*
(l) *f is strictly increasing.*
(m) *f possesses a maximum.*
(n) *f does not vanish on $\mathbb{R}$.*
(o) *f is not constant.*
(p) *f never has the same values at two different points.*

2. *Using the syntax of the predicates write the following sentences and then give their negation.*
   (a) *The square of any real number is positive or zero.*
   (b) *Any positive integer is the sum of three squares.*
   (c) *For any real number, we can find a real number such that their product is greater than 1.*
   (d) *Any nonempty part of $\mathbb{N}$ admits a smallest element.*

Exercise 5. *
Say if the following statements are true or false and write their negation.

1. $\forall x \in \mathbb{R}, x < 3 \Longrightarrow x^2 < 9.$
2. $\forall x, y \in \mathbb{R}, x < y \iff x^2 < y^2.$
3. $\exists a \in \mathbb{R}, \forall \varepsilon > 0, |a| < \varepsilon.$
4. $\forall a, b \in \mathbb{Z},\ a + ab + b = 0 \iff a = b = 0.$
5. $\exists x \in \mathbb{R},\ (x^3 + 3 = 0 \wedge x + 3 = 0).$
6. $(\exists x \in \mathbb{R},\ x^3 + 3 = 0) \wedge (\exists x \in \mathbb{R},\ x + 3 = 0).$
7. $\forall x, y \in \mathbb{R},\ \exists n \in \mathbb{N},\ nx > y.$
8. $\forall x \in \mathbb{R}_+^*,\ \forall y \in \mathbb{R},\ \exists n \in \mathbb{N},\ nx > y.$
9. $\forall x \in \mathbb{R},\ \exists m \in \mathbb{Z},\ m \leq x < m + 1.$
10. $\forall x \in \mathbb{R}_+^*,\ \forall y \in \mathbb{R},\ \exists n \in \mathbb{Z},\ 0 \leq y - nx < x.$
11. $\forall x \in \mathbb{R}_+^*, \forall y \in \mathbb{R}, \exists m \in \mathbb{Z} : m+1 \leq y - mx < m.$
12. $\forall x \in \mathbb{Z},\ \exists y \in \mathbb{Z},\ \forall z \in \mathbb{Z},\ z = x - y.$
13. $\exists x \in \mathbb{Z},\ \forall y \in \mathbb{Z},\ \forall z \in \mathbb{Z},\ z = x - y.$
14. $\forall x \in \mathbb{R}, \forall y \in \mathbb{R}, x^2 + y^2 \in \mathbb{R}_+.$
15. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x^2 - y^2 \in \mathbb{R}_+.$
16. $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x^2 - y^2 \in \mathbb{R}_+.$
17. $\forall A \subset E, \exists B \subset E, A \cup B = E.$
18. $\exists A \subset E, \forall B \subset E, A \cup B = E.$

Exercise 6. *

1. *Prove that*
   (a) $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$
   (b) $\displaystyle\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}.$
   (c) $\displaystyle\sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{n}{n+1}.$
   (d) $\displaystyle\sum_{k=1}^{n} \frac{1}{\sqrt{k}} < 2\sqrt{n}.$
   (e) $\forall n \in \mathbb{N},\ 2^{n-1} \leq n! \leq n^n.$
   (f) $\forall n \in \mathbb{N},\ a^n - b^n = (a-b)\displaystyle\sum_{k=0}^{n-1} a^k b^{n-1-k}$, *where* $a, b \in \mathbb{C}$.

2. *Prove each of the following statements using induction.*
   (a) *If $x > -1$ is any real number, then $(1+x)^n \geq 1 + nx$ for all $n \in \mathbb{N}$.*
   (b) *The number of parts with k elements of a set with n elements is equal to* $\dfrac{n!}{k!(n-k)!}$.

(c) *For any* $x_1, x_2, \ldots, x_n \in [0,1] : \prod_{i=1}^{i=n}(1-x_i) \geq 1 - \sum_{i=1}^{i=n} x_i$.

(d) *Any integer* $n > 2$ *is either prime or has a prime divisor* p *such that* p *is less or equal to* $\sqrt{n}$.

3. *Using induction show that*

   (a) $\forall n \in \mathbb{N}, \exists (p,q) \in \mathbb{N}_0^2,\ n = 2^p(2q+1)$.

   (b) $\forall n \in \mathbb{N}_0,\ 10|(n^5 - n)$.

   (c) *Every positive integer can be written as a product of prime factors, and this product is unique up to reordering of the factors.*

Exercise 7.

For any integer $n \geq 2$, *let* $H_n := \sum_{k=1}^{n} \frac{1}{k}$. H_n *is called the* n*-th harmonic number. The goal of this exercise is to show that* H_n *is not an integer for* $n \geq 2$.

1. *Show that for any positive integer* m, *there exist a positive integer* a *and nonnegative integer* b *such that* $H_{2m} = \frac{1}{2}H_m + \frac{a}{2b+1}$.
2. *Using induction, show that for any integer* $n \geq 2$, H_n *is a quotient of an odd integer and an even integer.*
3. *Deduce that* H_n *is not an integer for* $n \geq 2$.

Exercise 8. *

1. *Using the proof by contrapositive, show that the following statements are true.*

   (a) *For* $n \in \mathbb{N}$, *if* $3n^2 + n + 2$ *is odd then* n *is odd.*

   (b) *If* a^2 *is not an integer multiple of* 16, *then* $a/2$ *is not an even integer.*

   (c) *For two positive integers* a *and* n, *if* $a^n - 1$ *is a prime number, then* $a = 2$ *and* n *is a prime number.*

   (d) $\forall x \in \mathbb{R}, \forall y \in \mathbb{R}$, *if* $xy \neq 1$ *and* $x \neq y$, *then* $\frac{x}{x^2+x+1} \neq \frac{y}{y^2+y+1}$.

2. *Using the proof by contradiction, show that the following statements are true.*

   (a) $\forall n \in \mathbb{N}, \sqrt{n^2+1} \notin \mathbb{N}$.

   (b) *If* $a, b \in \mathbb{Z}$, *then* $a^2 - 4b - 2 \neq 0$.

   (c) *If* $a, b, c \in \mathbb{Z}$ *such that* $a^2 + b^2 = c^2$, *then* a *or* b *is even.*

   (d) $\sqrt{p} \notin \mathbb{Q}$, *for any prime number* p.

   (e) $\sqrt{2} + \sqrt{3} \notin \mathbb{Q}$.

   (f) $\frac{\ln 2}{\ln 3} \notin \mathbb{Q}$.

   (g) $\cos(\frac{\pi}{180}) \notin \mathbb{Q}$.

   (h) *Let* $p_1, p_2, \ldots, p_r$ *be prime numbers. The integer* $N = p_1 p_2 \ldots p_r + 1$ *is not divisible by any of the integers* p_i, $1 \leq i \leq r$.

Exercise 9.

Find two irrational numbers a, b *such that* a^b *is a rational number.*

Exercise 10. *

1. *Find all maps* $f : \mathbb{N}_0 \longrightarrow \mathbb{N}_0$ *that satisfy,* $f(n+m) = f(n) + f(m)$ *for all nonnegative integers* n *and* m*.*
2. *Find all maps* $g : \mathbb{N}_0 \longrightarrow \mathbb{N}_0$ *that satisfy,* $g(n+m) = g(n) \times g(m)$ *for all nonnegative integers* n *and* m*.*
3. *Find all functions* $h : \mathbb{R} \longrightarrow \mathbb{R}$ *that satisfy, for all* $x, y \in \mathbb{R}$*,* $h(y - h(x)) = 2 - x - y$*.*